Wireless and Mobile Computing

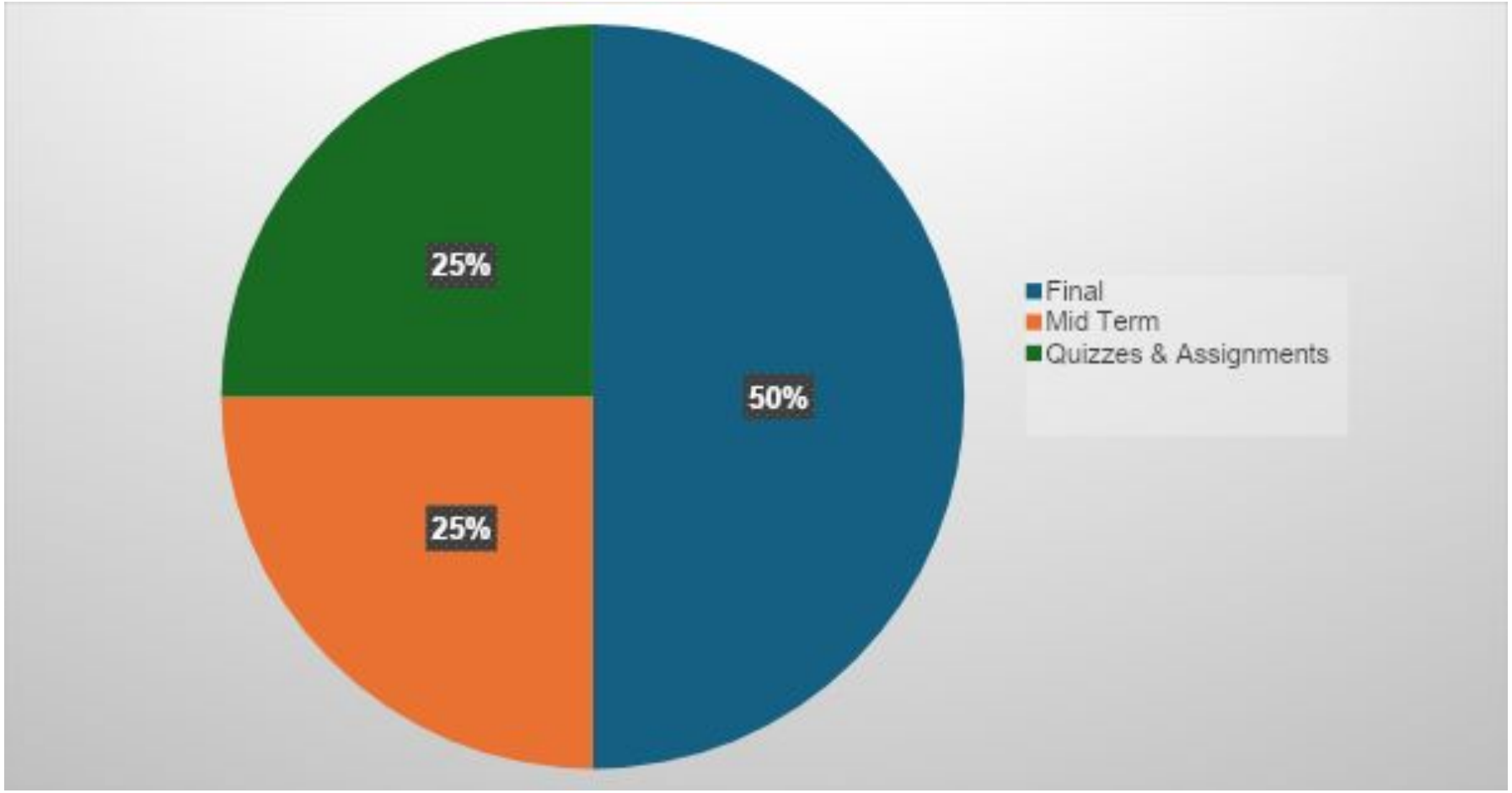
Lecture 1

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References

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3. Wireless Communication Networks and Systems, William Stallings.
4. Introduction to Wireless Communications and Networks A Practical Perspective, Krishnamurthy Raghunandan.
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Grading Policies



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Introduction

- In communication engineering, "**wireless**" refers to communication without the use of wires over any distance from a millimeter to vast deep space, beyond our solar system.

- A **wireless communication system** consists of several fundamental components that work together to transmit and receive data without physical connections. The basic components include:

1. Transmitter

•Converts the input data (audio, video, text, etc.) into an electromagnetic (radio) signal.

•Includes key elements such as:

- **Source Encoder**: Compresses data for efficient transmission.
- **Modulator**: Converts data into a signal suitable for wireless transmission.
- **RF Amplifier**: Boosts the signal strength for transmission.
- **Antenna**: Radiates the signal into space.

2. Channel (Wireless Medium)

•The medium through which signals travel, such as:

- Free space (air)
- Fiber optics (hybrid systems)
- Satellite links

•Affected by interference, noise, fading, and obstacles.

3. **Receiver**

•Captures and processes the transmitted signal.

•Includes:

- **Antenna**: picks up the transmitted signal.
- **RF Amplifier**: Boosts weak received signals.
- **Demodulator**: Extracts the original data from the signal.
- **Source Decoder**: Reconstructs the data from the received signal.

4. **User Equipment (End Devices)**

•Devices that interact with the wireless system, such as:

- Smartphones
- Laptops
- IoT devices
- Radios

5. **Base Station / Access Point**

•Acts as an intermediary between the transmitter and receiver.

•Examples:

- **Cell Towers** in mobile networks.
- **Wi-Fi Routers** in local networks.

6. **Control Systems & Protocols**

•Manage communication processes, including:

- **MAC (Medium Access Control) Layer**: Regulates access to the channel.
- **Network Protocols**: Define how data is transmitted (e.g., TCP/IP, LTE, 5G NR).
- **Error Control & Encryption**: Ensures reliable and secure data transmission.

Wireless Channel

- In a data transmission system, the **transmission medium** is the physical path between transmitter and receiver. Transmission media can be classified as guided or unguided.

- In both cases, communication is in the form of electromagnetic waves. With **guided media**, the waves are guided along a solid medium, such as copper twisted pair, copper coaxial cable, or optical fiber.
- The atmosphere and outer space are examples of **unguided media**, which provide a means of transmitting electromagnetic signals but do not guide them; this form of transmission is usually referred to as **wireless transmission.**
- Thus, a **wireless channel** is the medium through which electromagnetic waves propagate between a transmitter and a receiver in **a wireless communication system or without a physical path.**

- The characteristics and quality of a data transmission are determined both by the characteristics of the medium and the characteristics of the signal.
- In which type of transmission the signal's characteristics are usually more important than the medium characteristics?

Characteristics

Path Loss

Definition: The gradual reduction of signal power as it travels through the wireless medium.

Causes:

- Free-space loss (spreading of waves)
- Absorption by buildings, trees, and the atmosphere
- diffraction, and scattering

Effects:

- Reduces signal strength
- Limits communication range
- Requires power control or amplifiers to maintain connection

Example:

- A Wi-Fi router in one room has a weaker signal in another room due to **walls and obstacles**.

Fading

Definition: Rapid fluctuations in received signal strength over time or space due to changes in the transmission environment.

Effects:

- Causes **temporary signal drops** and distortions.
- Affects **voice calls and data transmission** in mobile networks.

Example:

- A phone call in a moving car **drops** when entering a tunnel due to sudden fading.

Interference

Definition: Unwanted signals from other electronic devices or networks that degrade communication quality.

Sources of Interference:

- **Co-Channel Interference (CCI)**
  - Occurs when multiple devices operate on the same frequency channel.
  - Example: Multiple Wi-Fi routers using the same channel in an apartment building.
  - Solution: Use channel planning to reduce overlap.
- **Adjacent-Channel Interference (ACI)**
  - Happens when signals from nearby channels overlap due to improper filtering.
  - Example: Two Wi-Fi networks operating on channels that are too close (e.g., 1 and 2).
  - Solution: Use non-overlapping channels (1, 6, 11 in 2.4 GHz Wi-Fi).
- **Electromagnetic Interference (EMI)**
  - Caused by electronic devices emitting radio waves that interfere with wireless signals.
  - Example: Microwaves, Bluetooth devices, cordless phones interfering with Wi-Fi.
  - Solution: Minimize the use of such devices near Wi-Fi routers.

Effects:

- Reduces signal quality and network performance.
- Causes **packet loss, slow speeds, and connection drops**.

Multipath Propagation

Definition: The phenomenon where a transmitted signal takes **multiple paths** to reach the receiver due to reflections, scattering, and diffraction.

Causes:

- **Reflection:** Signal bounces off buildings, walls, and surfaces.
- **Scattering:** Signal deflects in different directions due to small objects.
- **Diffraction:** Signal bends around obstacles (e.g., buildings).

Effects:

- **Constructive Interference:** Improves signal strength.
- **Destructive Interference:** Causes fading and distortion.

Doppler Effect

Definition: The change in frequency of a signal due to the motion of the transmitter, receiver, or obstacles.

Causes:

- User moving with respect to the base station (e.g., in a car or train).
- Moving obstacles (e.g., passing vehicles, airplanes).

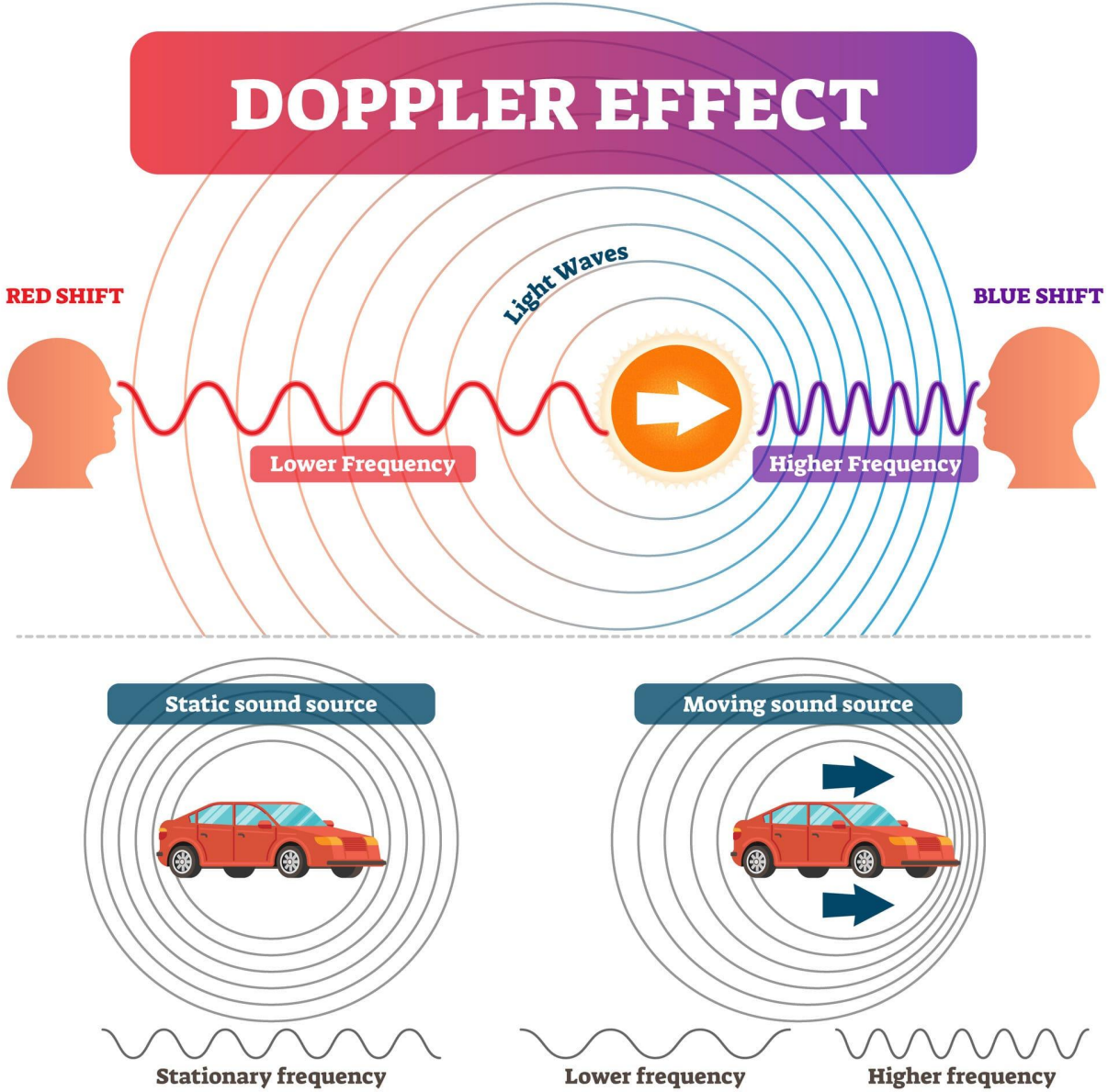
Effects:

- Frequency shifts lead to **distortion** in signal reception.
- Reduces **data transmission accuracy** in high-speed environments.

Example:

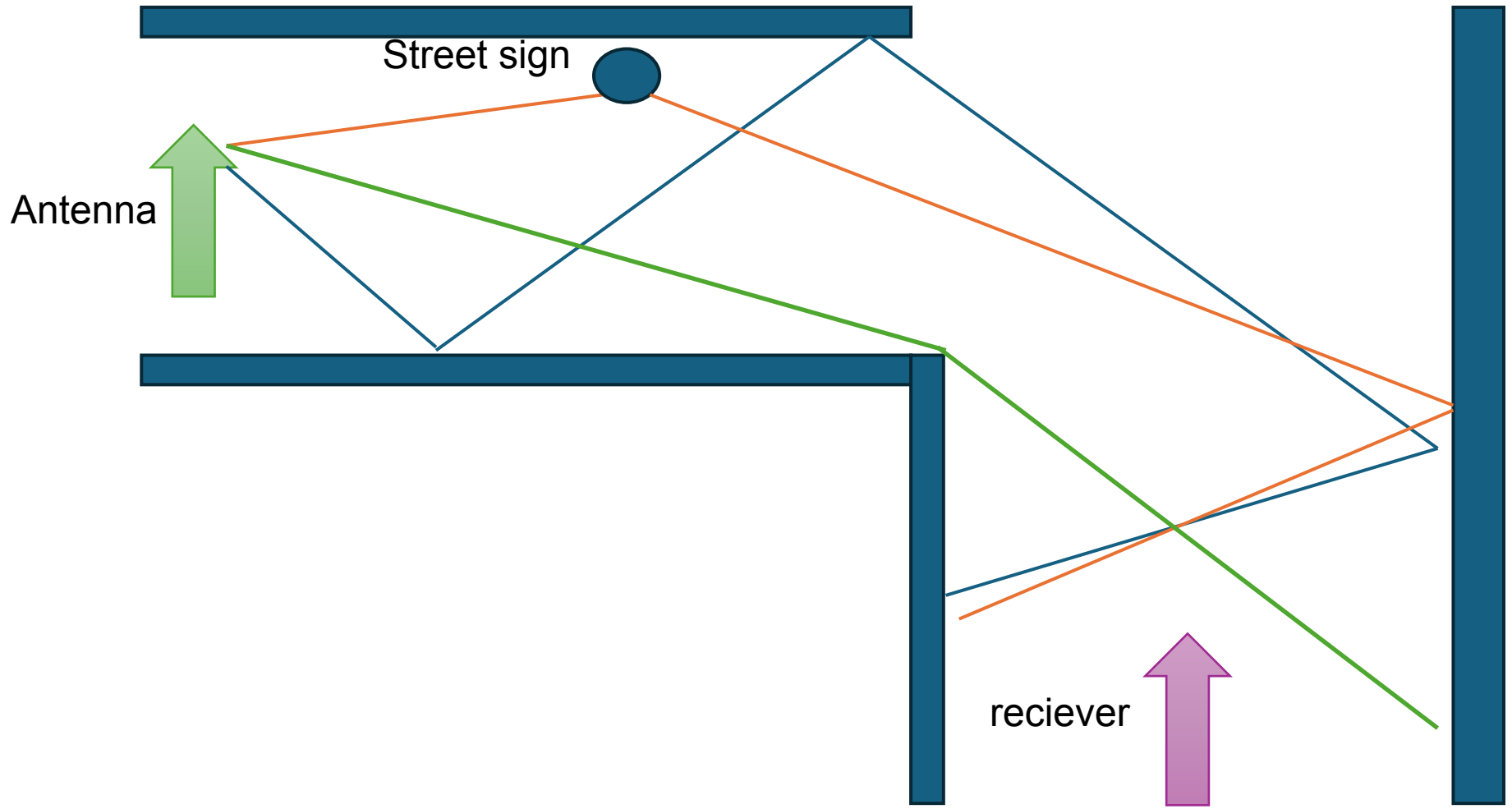
- FM radio signal changes when driving fast due to **frequency shift**.



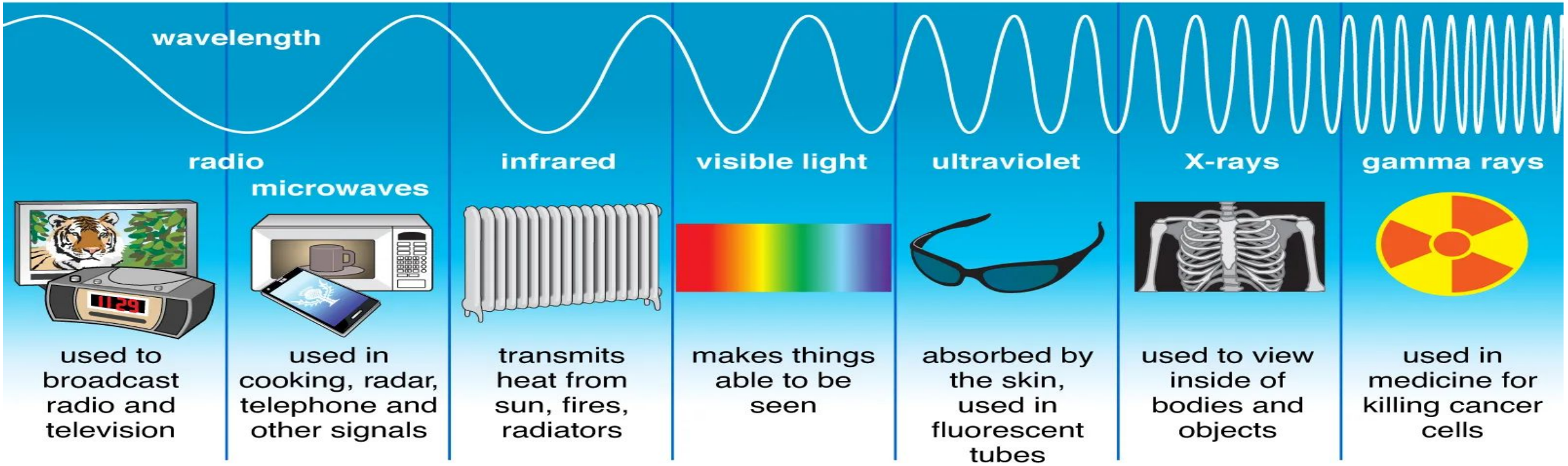

DOPPLER EFFECT
RED SHIFT
Light Waves
BLUE SHIFT
Lower Frequency
Higher Frequency
Static sound source
Moving sound source
Stationary frequency
Lower frequency
Higher frequency

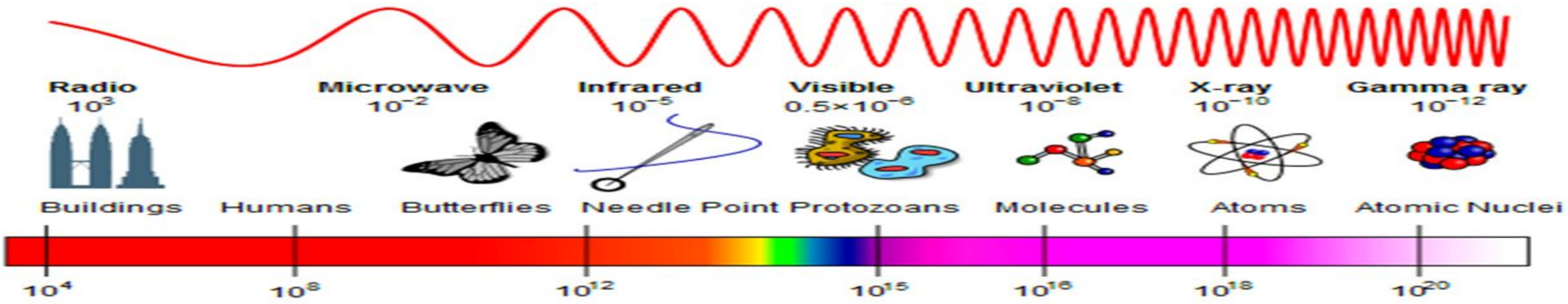
Mechanisms of electromagnetic propagation

1. **Line of sight**
2. **Reflection ($d \gg \lambda$)**
3. **Diffraction (with sharp edges)**
4. **Scattering ($d \ll \lambda$)**



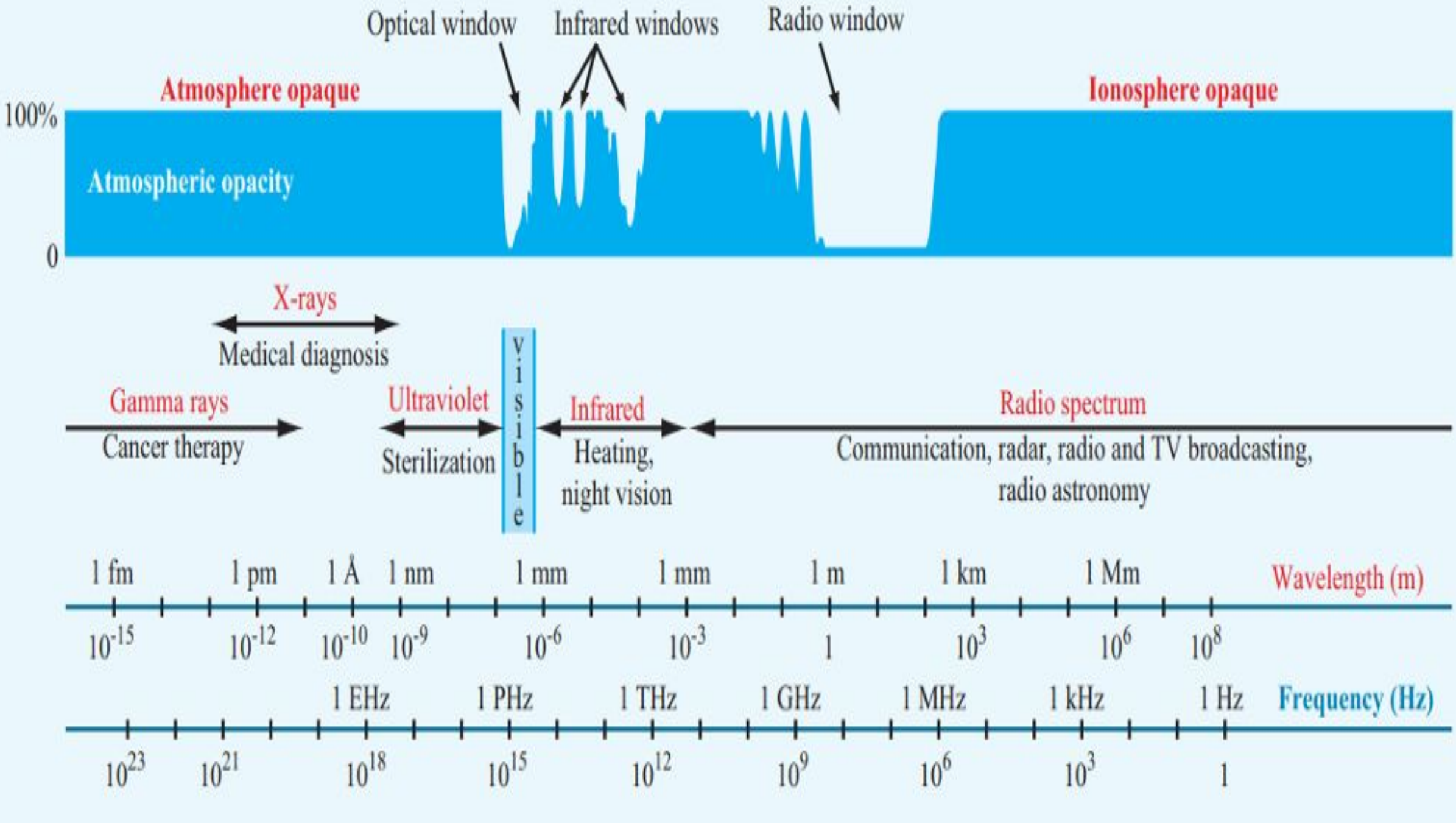
Types of Electromagnetic Radiation

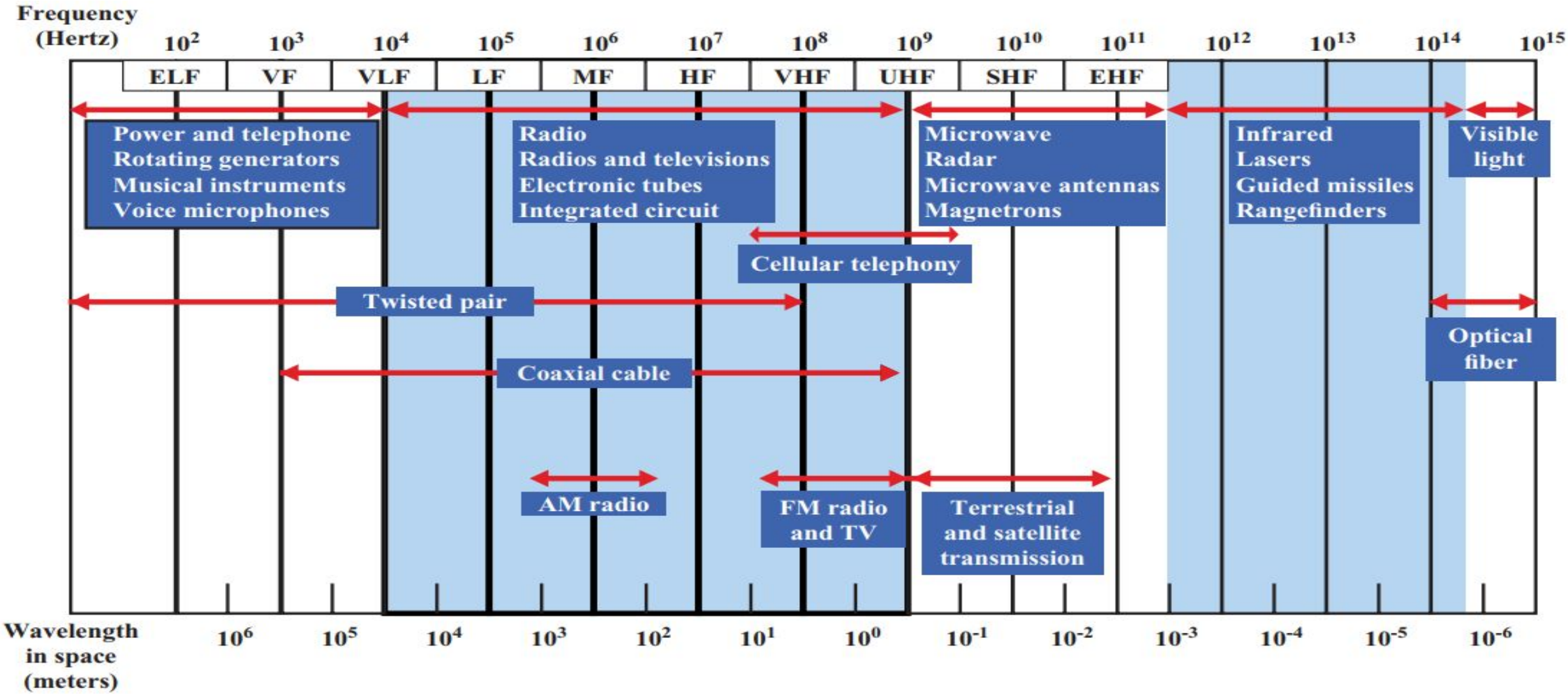




Atmospheric windows to EM radiation

- Earth's atmosphere acts as an opaque barrier to much of the electromagnetic spectrum.
- The atmosphere absorbs most of the wavelengths shorter than ultraviolet, most of the wavelengths between infrared and microwaves, and most of the longest radio waves.
- Three general ranges of frequencies are of interest in wireless transmission. Frequencies in the range of about 300 MHz to 300 GHz are referred to as microwave frequencies.
- Frequencies in the range of 3 KHz to 300 GHz are referred to as the radio range.
- The infrared portion of the spectrum covers, roughly, from 300 GHz to 400 THz.
- As atmospheric conditions change the radio window can expand or shrink.


Optical window
Infrared windows
Radio window
Atmosphere opaque
Ionosphere opaque
100%
Atmospheric opacity
0
X-rays
Medical diagnosis
Gamma rays
Cancer therapy
Ultraviolet
Sterilization
visible
Infrared
Heating, night vision
Radio spectrum
Communication, radar, radio and TV broadcasting, radio astronomy
1 fm
1 pm
1 Å
1 nm
1 mm
1 mm
1 m
1 km
1 Mm
Wavelength (m)
10^{-15}
10^{-12}
10^{-10}
10^{-9}
10^{-6}
10^{-3}
1
10^{3}
10^{6}
10^{8}
1 EHz
1 PHz
1 THz
1 GHz
1 MHz
1 kHz
1 Hz
Frequency (Hz)
10^{23}
10^{21}
10^{18}
10^{15}
10^{12}
10^{9}
10^{6}
10^{3}
1



ELF = Extremely low frequency
VF = Voice frequency
VLF = Very low frequency
LF = Low frequency
MF = Medium frequency
HF = High frequency
VHF = Very high frequency
UHF = Ultrahigh frequency
SHF = Superhigh frequency
EHF = Extremely high frequency

Figure 2.10 Electromagnetic Spectrum for Telecommunications

Feature	Radio Waves	Microwaves	Infrared (IR)
Frequency Range	3 kHz – 300 GHz	300 MHz – 300 GHz	300 GHz – 400THz
Wavelength range	1 mm – 100 km	1 mm – 1 m	700 nm – 1 mm
Penetration Ability/ material	Penetrates walls, buildings, and non-metallic objects	Limited penetration through walls, absorbed by rain	Blocked by walls, fog, dust, and smoke
Propagation Mode	Ground wave, skywave, and line-of-sight	Line-of-sight and some reflection	Line-of-sight only
Interference	High (overlapping signals, noise, and weather effects)	Moderate (affected by water vapor and objects)	Low (affected by direct light sources)
Common Uses	AM/FM radio, TV broadcasting, mobile networks (cellular network), satellites	Wi-Fi, satellite, radar, 5G, microwave ovens	Remote controls, short-range communication, thermal imaging

Feature	Radio Waves	Microwaves	Infrared (IR)
Mobility	High (used for mobile communication)	Moderate (used in some mobile applications)	Low (used for fixed short-range communication)
Suitability for Mobile Communication	Best choice (widely used for cellular networks and Wi-Fi).	Good but limited (used in high-speed networks like 5G mmWave and satellite links).	Not suitable (short-range, requires line-of-sight)
Data Transmission Speed	Moderate (best for long-distance communication but slower than microwaves & IR for data.	High (used in high-speed wireless communication like Wi-Fi, 5G, and satellites.	Fastest speeds (fiber optics), but limited to short-range and line-of-sight.

Wave	Directional Property	Explanation
Radio Waves	Omnidirectional (in most cases).	Radio waves usually spread out in all directions unless focused by antennas (e.g., parabolic or Yagi antennas). This makes them ideal for broadcasting (e.g., FM radio, TV signals, and cellular networks).
Microwaves	Highly directional.	Microwaves are used in point-to-point communication, requiring highly focused antennas (e.g., satellite dishes, microwave links). Higher-frequency microwaves (e.g., millimeter waves) are even more directional.
Infrared (IR)	Strictly directional.	Infrared waves travel in a straight line and do not bend around obstacles. They require line-of-sight, which makes them ideal for remote controls, optical fiber communication, and infrared sensors.